Partition Repair Maps and Exact Weighted Congestion

21 July 2026

Abstract

We give the exact finite theorem needed to convert a geometric repair of bad configurations into a normalized weighted partition-numerator bound, retaining multiplicity, state weight, and Jacobian loss.

Let $T : B \to G$ map bad states to good states. For good weights g_y, bad weights b_x, present weight $V(s)$, and next-state weight $V(\tau x)$, define

$$C_T = \max_{y \in G} \frac{\sum_{x:T(x)=y} b_x V(\tau x)}{g_y V(s)}.$$

Theorem 1 (Exact repair congestion)**.** *The weighted bad numerator obeys*

$$B_V(s) \le C_T Z_G(s) V(s).$$

Thus the ratio to the good partition is at most C_T.

Proof. Group the bad sum by repair fibers and apply the definition of C_T to each good state before summing. □

Corollary 1. *If every repair fiber has at most M elements and each repaired density ratio, including state and Jacobian factors, is at most $e^{J-\Delta}$, then*

$$C_T \le M e^{J-\Delta}, \qquad F_T \ge \Delta - J - \log M.$$

Exponential repair multiplicity can cancel an action drop, reproducing the mandatory entropy counterexample.

Remark 1. *The theorem gives the exact normalization interface. A nonlinear, gauge-covariant $SU(2)$ repair with uniform congestion remains open.*